

FOUNDATIONAL CONCEPTS

Project: a temporary group of related tasks undertaken to create a unique product, service, or result

Program: a group of related projects and operations coordinated together in a larger effort than a project

Portfolio: an organization's entire investment in projects and programs (see diagram 1)

Project Management: using skills, knowledge, and resources to satisfy project requirements

Process: 1 of 42 packages of inputs, tools, and outputs used to produce specific output(s) for the project (see diagram 2)

Phases: groups of project activities separated by exit gates to evaluate phase deliverables

Project Life Cycle: a group of project phases specified by an organization's project management methodology

Methodology: an organization's specific implementation of project processes; a set of steps to manage a project

Triple Constraint: the concept that scope, time, and cost are interrelated; one cannot change without affecting at least one of the others; also known as the "Iron Triangle" (see diagram 3)

Baseline (BL): the original plan plus all approved changes (scope, schedule, cost)

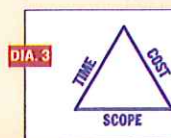
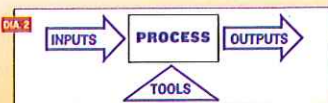
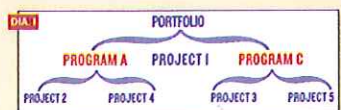
System: a set of formal management procedures; rules, processes, people, etc. that support an outcome

Historical Information: information from previous projects used to evaluate future project decisions

Lessons Learned: documented variances (historical information) used to avoid variances in future projects

Progressive Elaboration: an iterative approach where planning occurs in cycles rather than only up front

PM Skills: Leading, Communicating, Negotiating, Problem-solving, Influencing, Delegating, and being Proactive



PROJECT ROLES

Project Manager: the person ultimately responsible for the project; authorized to spend budget and assign resources to realize project goals

Project Coordinator: a role weaker than a project manager; may assign project resources but not authorized to spend funds

Project Expeditor: a role weaker than a project coordinator; reports on project progress without authority to assign resources or spend funds

Project Sponsor: responsible for funding the project; typically a senior manager; may or may not be the customer

Senior Management: a role higher in organization than the project manager; prioritize projects, authorize PM, resolve organization conflicts and issues

Stakeholder: a person or group having an interest, whether positive or negative, in the result of the project

Functional Manager: a department manager who "owns" project resources; administers human resource management responsibilities for employees

Project Office: a supporting or controlling department regarding methodologies, tools, standards, best practices, and audits

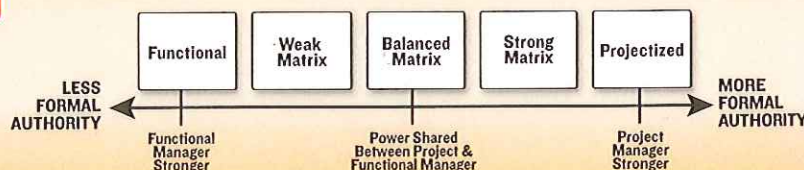
Program Manager: responsible for programs, coordinating several related projects to achieve a common goal

ORGANIZATION TYPES & PROJECT MANAGER POWER

Functional: team members work for a department; may be loaned to a project at times; little or no PM power

Projectized: PM is manager of projects and resources; has complete authority

Matrix: Combination with functional managers and project managers; PM has more power in Strong Matrix, equal in Balanced Matrix, and less in Weak Matrix



PROCESS FRAMEWORK

Process groups are not project phases. Process groups are not linear; they are iterative. Process names follow verb-noun structure (e.g., Define Scope).

KNOWLEDGE AREAS (9) (# processes)

Integration Management (INT)	6
Scope Management (SCO)	5
Time Management (TIM)	6
Cost Management (COS)	3
Quality Management (QUA)	3
Human Resource Management (HUM)	4
Communications Management (COM)	5
Risk Management (RIS)	6
Procurement Management (PRO)	4

42 PROCESSES

MATRIX	IN	PL	EX	MC	CL
INT	1	1	1	2	1
SCO		3		2	
TIM		5		1	
COS		2		1	
QUA		1	1	1	
HUM		1	3		
COM	1	1	2	1	
RIS		5		1	
PRO		1	1	1	1

PROCESS GROUPS (5) (# processes)

Initiating (IN)	2
Planning (PL)	20
Executing (EX)	8
Monitoring and Controlling (MC)	10
Closing (CL)	2

COMMON INPUTS, TOOLS, AND OUTPUTS

Inputs: Enterprise Environmental Factors (EEFs), Organizational Process Assets (OPAs), Project Management Plan (PM Plan), Work Performance Information (WPI)

Tools: Expert Judgment, Project Management Information System (PMIS)

Outputs: Change Requests, Updates (to PM Plan, Project Documents, and OPAs)

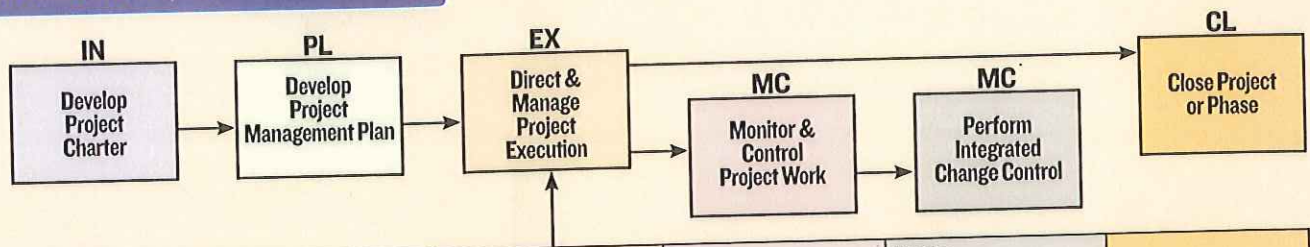
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THE PMP® EXAM
HOW TO PASS ON YOUR FIRST TRY

by Andy Crowe

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INTEGRATION MANAGEMENT (INT)



KEY:

INPUTS	Project Statement of Work Business Case	Project Charter Planning Outputs	PM Plan Approved Change Requests	PM Plan Performance Reports	PM Plan Work Performance Information Change Requests	PM Plan Accepted Deliverables
TOOLS	Expert Judgment	Expert Judgment	Expert Judgment Proj Mgmt Info System	Expert Judgment	Expert Judgment Change Control Meetings	Expert Judgment
OUTPUTS	Project Charter	PM Plan	Deliverables Work Performance Information	Change Requests Plan & Doc Updates	Change Request Status Updates	Final Transition

NOTES ABOUT INTEGRATION MANAGEMENT

Business Case: Market demand, business need, customer requirement, technology advance, legal requirement, ecological impact, social need

Project Selection Methods:

(Bigger is Better) Benefit-cost ratio, economic value add (Net Operating Profit After Tax - Cost of Capital), internal rate of return, present value, net present value, return on investment, return on invested capital (Net Income After Tax - Invested Capital) (Smaller is Better) opportunity cost, payback period

Project Charter: Names & authorizes PM to expend resources to achieve project objectives (may include high-level requirements, milestones, and budget)

PM Plan: Formal, approved document defining how project is managed, executed, and controlled (There are 15 components, see table at right.)

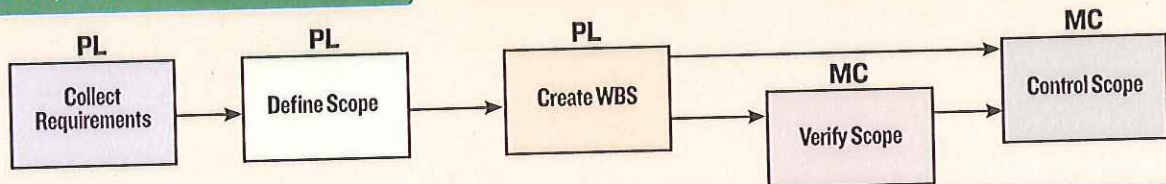
Work Authorization System: Part of the overall PMIS; ensures that project work gets done at the right time and in the right sequence; may be as informal as an email or as formal as a system to assign and log job work orders

Change Requests: Approved or rejected in Perform Integrated Change Control; PM should influence factors that cause change; PM should assess impact of changes on project and make decision

Close Project or Phase: This is the last process to be completed in each phase or in the project; capture lessons learned, archive project documents, celebrate success, and release project team

PM PLAN COMPONENT	PROCESS WHERE CREATED
Change Management Plan	Develop Project Management Plan
Configuration Management Plan	
Scope Management Plan	
Schedule Management Plan	
Cost Management Plan	
Quality Management Plan	Plan Quality
Process Improvement Plan	
Human Resource Plan	Develop Human Resource Plan
Communications Management Plan	Plan Communications
Risk Management Plan	Plan Risk Management
Procurement Management Plan	Plan Procurements
Requirements Management Plan	Collect Requirements
Scope Baseline	Create WBS
Schedule Baseline	Develop Schedule
Cost Performance Baseline	Determine Budget

SCOPE MANAGEMENT (SCO)



KEY:

INPUTS	Project Charter Stakeholder Register	Project Charter Requirements Documentation	Project Scope Statement Requirements Documentation	Validated Deliverables PM Plan (Scope Baseline)	PM Plan (Scope Baseline) Work Perf Information
TOOLS	Facilitated Workshops Group Creativity Technology Group Decision-making	Facilitated Workshops Alternatives Identification Product Analysis	Decomposition	Inspection	Variance Analysis
OUTPUTS	Requirements Documentation Requirements Traceability Matrix	Project Scope Statement	WBS & WBS Dictionary Scope Baseline (BL)	Accepted Deliverables Change Requests	Work Performance Measurements Change Requests

NOTES ABOUT SCOPE

Group Creativity Techniques: Brainstorming, Nominal group technique, Delphi technique, Idea and mind mapping

Delphi Technique: Gathering expert opinion without participants knowing who else is being polled; prevents biasing opinions and groupthink

Nominal Group Technique: Brainstormed ideas are voted upon and ranked by priority

Group Decision-making Techniques: Unanimity, majority, consensus, plurality, and dictatorship

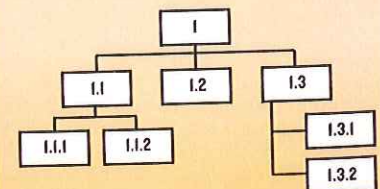
Project Scope Statement: Document stating the project requirements by describing objectives, deliverables, boundaries, and acceptance criteria

Decomposition: Breaking down scope statement (project deliverables) into progressively smaller pieces

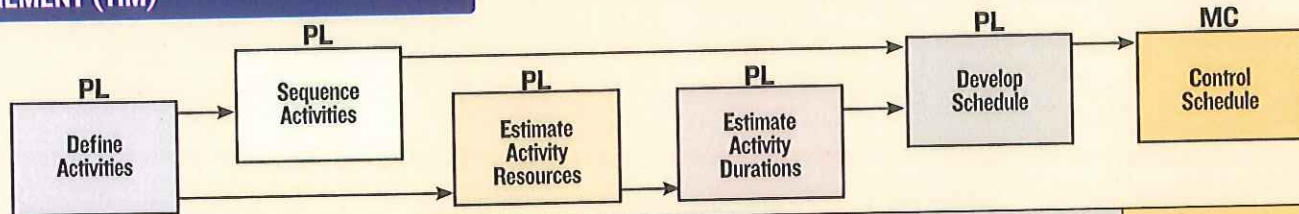
WBS (Work Breakdown Structure): An hierarchical decomposition of deliverables to increasingly more detail for better estimating, assignment, and tracking; each node has a unique number; Work Packages are the lowest level nodes, decomposed low enough to be estimated (cost & time) and assigned (see diagram at right).

Scope Baseline: A combination of the original WBS, WBS dictionary, and project scope statement plus all approved changes

Work Breakdown Structure (WBS)



TIME MANAGEMENT (TIM)



KEY:

INPUTS	Scope Baseline	Activity List/Attributes Project Scope Statement Milestone List	Activity List/Attributes Resource Calendars	Activity List/Attributes Project Scope Statement Resource Calendars Activity Resource Requirements	Activity List/Attributes Project Scope Statement Resource Calendars Act. Res. & Duration Requirements Project Network Diagrams	Project Schedule PM Plan (Schedule BL) Work Perf Information
TOOLS	Decomposition Rolling Wave Planning	Precedence Diagramming Method Dependency Determination Apply Leads & Lags	Expert Judgment Alternatives Analysis Published Data Bottom-Up Estimating	Analogous (Top-Down) Estimating Parametric Estimating Reserve Analysis Three-Point Estimates	Critical Path / Critical Chain Resource Leveling Schedule Compression Apply Leads & Lags What-If Analysis Sched Network Analysis	Performance Reviews Resource Leveling Schedule Compression Adjust Leads & Lags What-If Analysis Variance Analysis
OUTPUTS	Activity List/Attributes Milestone List	Project Schedule Network Diagrams	Activity Resource Requirements Resource Breakdown Structure	Activity Duration Estimates	Project Schedule Schedule Baseline	Work Performance Measurements Change Requests

NOTES ABOUT TIME MANAGEMENT

Project Network Diagram: a diagram of the schedule activities in the order in which they must be performed; may be AON or AOA (see diagram 4)

Precedence Diagramming Method (PDM): network diagram where activities are represented by the nodes (rectangles) and the arrows are dependencies; also called AON (Activity-On-Arrow)

AOA (Activity-On-Arrow): network diagram where activities are represented by the arrows connecting nodes; the nodes are points in time; AOA may feature "dummy activities" which are actually dependencies, shown as dashed arrows with 0 duration; this method also called Arrow Diagramming Method.

Types of Dependencies: Mandatory (hard logic), Discretionary (preferred or soft logic), and External (beyond or outside of project control)

Logical Relationships: dependencies between two activities where one activity must be started or finished before another activity can be started or finished; Four types: Finish-to-Start, Finish-to-Finish, Start-to-Finish, Start-to-Start (e.g., F-S: Task A must finish before Task B can start)

Lead: starting an activity prior to the finish of a preceding activity

Lag: delaying the start of an activity after the finish of a preceding activity

Estimating, Analogous: (also called top-down) using historical information from similar projects/activities to estimate effort, duration, or cost.

Estimating, Bottom-up: estimating effort, duration, or costs at lowest levels of schedule or WBS and aggregating up to summary WBS nodes.

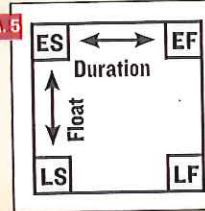
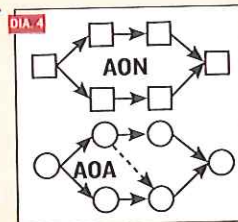
Estimating, Parametric: estimating effort, duration, or costs using past performance information; best on scalable or linear components; unit time or cost * number of units; e.g. 4 resources are required to clear 1 acre in 1 day, therefore 16 resources could clear 4 acres in 1 day.

Estimating, Three-Point: weighted average of effort, duration, or cost (also called PERT estimate); Formula: $(\text{Pessimistic} + 4 \times \text{Realistic} + \text{Optimistic}) \div 6$; the standard deviation of this probabilistic estimate is calculated as $(\text{Pessimistic} - \text{Optimistic}) \div 6$. (A statistical short-cut)

Network Path: a series of schedule activities having a relationship that carries through from the project's start to finish.

Critical Path: the path(s) of schedule activities where the delay of any one activity would delay the project finish; the path of highest risk.

Float (Slack): the amount of time a schedule activity could be delayed without delaying the project finish (see diagram 5)



Free Float: the amount of time a schedule activity could be delayed without delaying the early start (ES) of a subsequent activity (see diagram 5)

Negative Float: the situation of a schedule activity's early finish (EF) being after a subsequent activity's early start (ES) (see diagram 5)

Critical Path Method (CPM): schedule analysis to determine the critical path, overall schedule, and each activity's float

Critical Chain Method (CCM): aggressive schedule management based on managing schedule buffers and keeping resources fully applied (Eliyahu Goldratt)

Schedule Compression: Crashing (adding resources & cost) or Fast Tracking (adjusting dependencies and increasing risk) to shorten the overall schedule

Resource-leveling: Adjusting required resources to the level of resources available to the organization (usually lengthens schedule)

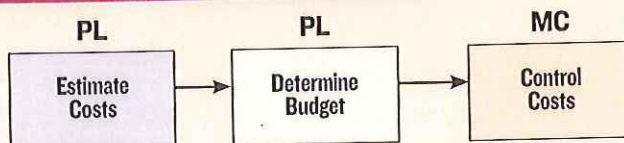
What-if Analysis: Monte Carlo analysis/simulation - compute large numbers of possible scenarios related to schedule (also used in Risk Management)

Milestone Schedule: a high-level schedule showing only significant schedule points

Gantt Chart (bar chart): a schedule using horizontal bars to represent activities; the bar length represents activity duration; placement represents start and finish dates.

Schedule Baseline: a formal version of the project schedule that is placed under control; the schedule baseline is a component of the project plan.

COST MANAGEMENT (COS)



KEY:

INPUTS	Scope Baseline Project Schedule Risk Register Human Resource Plan	Activity Cost Estimates Project Schedule Resource Calendars	PM Plan (Cost Baseline) Project Funding Requirements Work Performance Information
TOOLS	Analogous Estimate (Top-Down) Parametric Estimating Bottom-Up Estimating Three-Point Estimates Reserve Analysis Cost of Quality Vendor Bid Analysis	Cost Aggregation Reserve Analysis Funding Limit Reconciliation	Earned Value Measurement Forecasting Variance Analysis TCPI (To-Complete-Performance-Index) Performance Reviews
OUTPUTS	Activity Cost Estimates	Cost Performance Baseline Project Funding Requirements	Work Performance Measurements Budget Forecasts Change Requests

EARNED VALUE FORMULAS

BAC	Budget At Completion	Total budget
PV	Planned Value (BCWS)	Planned % Complete * BAC
EV	Earned Value (BCWP)	Actual % Complete * BAC
AC	Actual Cost (ACWP)	Sum of actual costs
CV	Cost Variance	EV - AC
SV	Schedule Variance	EV - PV
SPI	Schedule Performance Index	EV ÷ PV
CPI	Cost Performance Index	EV ÷ AC
EAC	Estimate At Completion	BAC ÷ CPI
ETC	Estimate To Completion	EAC - AC
VAC	Variance At Completion	BAC - EAC
TCPI	To-Complete-Perf-Index	(BAC - EV) ÷ (BAC - AC)

Variance: Difference (subtraction) between planned and actual; target value is 0; positive is good, negative is bad

Performance Index: Ratio (division) of planned to actual; target value is 1; greater than 1 is good, less than 1 is bad; it is never negative

Forecast: EAC, ETC, and VAC are all forecasted values

COST MANAGEMENT (COS) (CONTINUED...)

Life-cycle Costing: looking at total cost of ownership from purchase (creation) through operations to disposal

Value Engineering: getting the most project value in cost, quality, schedule, and benefit without sacrificing scope

Cost Aggregation: rolling up activity costs to the WBS node where they will be managed

Cost Performance Baseline (budget): what costs will be incurred when in the project (the S-curve is the cumulative sum of the budgeted costs over the project life; costs start slowly, accelerate, then taper off, forming an S pattern)

Funding Limit Reconciliation: adjusting project costs and timeframes to fit within funding availability timeframes; typically funding requirements differ from the cost performance baseline by the risk reserves

Cumulative: from the start of the project up to a point in time (e.g., CPI at the end of month 5, or total actual costs to date); sometimes notated using a superscript e.g., (CPI^c or AC^c); Cumulative CPI: $CPI^c = EV^c \div AC^c$

Periodic: opposite of cumulative; for a given period of time (e.g., CPI during month 2, or planned % complete in month 6)

Working Efficiency: SPI, schedule performance index, optimally 1

To-Complete-Performance-Index: spending efficiency required to meet targeted financial goals (usually greater than 1 if CPI is less than 1, or less than 1 if CPI is greater than 1); calculated (in terms of money) as remaining work (BAC-EV) divided by remaining funds (BAC-AC), when BAC is the targeted finish goal; if the targeted finish goal is EAC, then TCPI is calculated as $(BAC-EV) \div (EAC-AC)$

Variance Analysis: Measuring the difference between planned and actual (costs or durations); variances are optimally zero (on budget and/or on schedule)

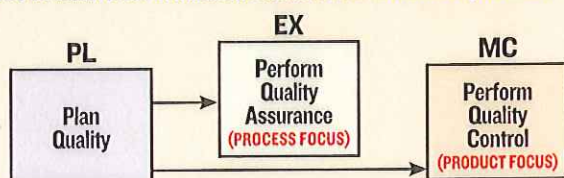
Spending Efficiency: CPI, cost performance index, optimally 1

NOTES ABOUT COST MANAGEMENT

Type of Estimate	Typical Range
Order of Magnitude	-50% to +100%
Conceptual	-30% to +50%
Preliminary	-20% to +30%
Definitive	-15% to +20%
Control	-10% to +15%

QUALITY MANAGEMENT (QUA)

QUALITY: the degree to which a set of inherent characteristics fulfills requirements



KEY:

INPUTS	Scope Baseline Schedule Baseline Cost Perf Baseline	Quality Metrics Work Perf. Information Quality Control Meas.	Metrics / Checklists Work Perf. Meas. Deliverables
TOOLS	Benchmarking Cost-Benefit Analysis Cost of Quality Control Charts Design of Experiments Flowcharting Statistical Sampling	Quality Audits Process Analysis Plan Quality TOOLS FROM Performance Quality Control	Inspection Fishbone Diagrams Control & Run Charts Flowcharting Pareto Charts Scatter Diagrams Statistical Sampling
OUTPUTS	Quality Mgmt. Plan Metrics / Checklists Process Improv. Plan	Change Requests	Validated Deliverables Quality Control Meas. Validated Changes

NOTES ABOUT QUALITY MANAGEMENT

Prevention Over Inspection: current quality philosophy based on the fact that it is less expensive to prevent an error than to fix one

TQM (Total Qual Mgmt): everyone in organization is responsible for quality

Kaizen (continuous improvement): constant process improvement

JIT (Just-In-Time): near-zero inventory; no spares

ISO 9000: certification that you document and follow processes

Six Sigma: 3.4 defects per million; 99.99966% meet specifications

Three Sigma: 2700 defects per million; 99.73% meet specifications

One Sigma: 317,500 defects per million; 68.25% meet specs.

Mutually Exclusive: one choice or event excludes other possibilities

Statistical Independence: 2 outcomes are not dependent on each other

Probability Distribution: table or graph showing mathematical possibility of an event occurring

Standard Deviation: a measure of diversity of data observations

Cost of Quality (COQ): quality planning and appraisal costs

Control Chart: graph of samples to determine if process is in control

Run Chart: graph of a statistic over time to depict change

Statistical Sampling: random sampling to cut costs of 100% sample

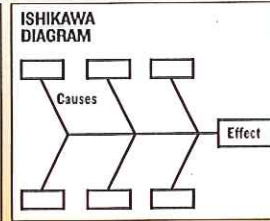
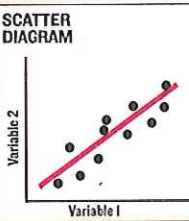
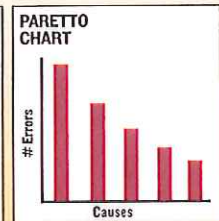
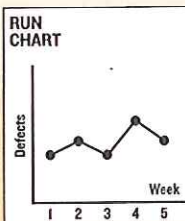
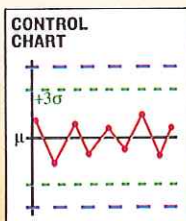
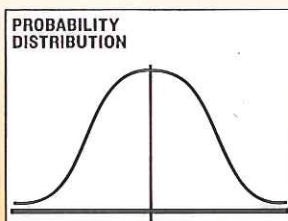
Benchmarking: comparing project quality standards to those of other projects

Flowcharting: depicting relationships of components in a process

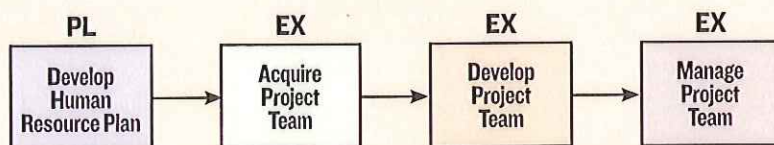
Pareto Chart: depicting causes descendingly ranked by number of problems resulting from the respective causes

Scatter Diagram: plot of data points against 2 variables to depict possible statistical correlation or dependence of the variables

Cause & Effect (Ishikawa/Fishbone) Diagram: relating potential underlying causes to an effect or result in analysis



HUMAN RESOURCE MANAGEMENT (HUM)



KEY:

INPUTS	Activity Resource Requirements	PM Plan (HR Plan)	Staff Assignments Resource Calendars	Staff Assignments PM Plan (HR Plan) Team Performance Assessment Performance Reports
TOOLS	Org Charts & Position Descriptions Networking Org Theory	Pre-assignment Negotiation Acquisition Virtual Teams	Interpersonal Skills Training Team-building Activities Ground Rules Co-location Recognition & Rewards	Interpersonal Skills Observation & Conversation Project Performance Appraisals Conflict Management Issue Log
OUTPUTS	Human Resource Plan	Staff Assignments Resource Calendars	Team Performance Assessments	Change Requests

MOTIVATIONAL THEORIES

Maslow's Hierarchy of Needs: 5 categories of needs exist in an hierarchy; meeting needs in one category does not motivate unless the lower level needs are already met

Herzberg's Motivation-Hygiene: hygiene factors do not motivate but must be present for motivational factors to function

McGregor's Theory-X & Theory-Y: X-mgrs believe constant supervision is necessary; Y-mgrs believe people can be trusted to work

Fiedler's Contingency: effectiveness of a leader's style (task- or relationship-oriented) is contingent on the situation

McClelland's Theory of Needs: workers may be motivated by meeting their varying degrees of need for (1) achievement, (2) power, or (3) affiliation

Expectancy: the realistic (believable) expectation of a reward is a motivator

HUMAN RESOURCE MANAGEMENT (HUM) CONTINUED...

RAM (Responsibility Assignment Matrix):
work packages in row headings and roles in column headings;
checked cells relate work to roles

RACI Chart: specific type of RAM; cells contain 'R'
Responsible, 'A' Accountable, 'C' Consult, or 'I' Inform
(relating role to work package)

Staffing Management Plan: contained in HR Plan;
how and when team will be staffed, trained, and released

CONSTRUCTIVE TEAM ROLES

Initiators
Information Seekers
Information Givers
Encouragers
Clarifiers
Harmonizers
Summarizers
Gate Keepers

DESTRUCTIVE TEAM ROLES

Aggressors
Blockers
Withdrawers
Recognition Seekers
Topic Jumpers
Dominators
Devil's Advocates

FORMS OF POWER

Reward (best)
Expert (best)
Legitimate
Referent
Punishment (worst)

INTERPERSONAL SKILLS

Leading
Influencing
Effective Decision-making

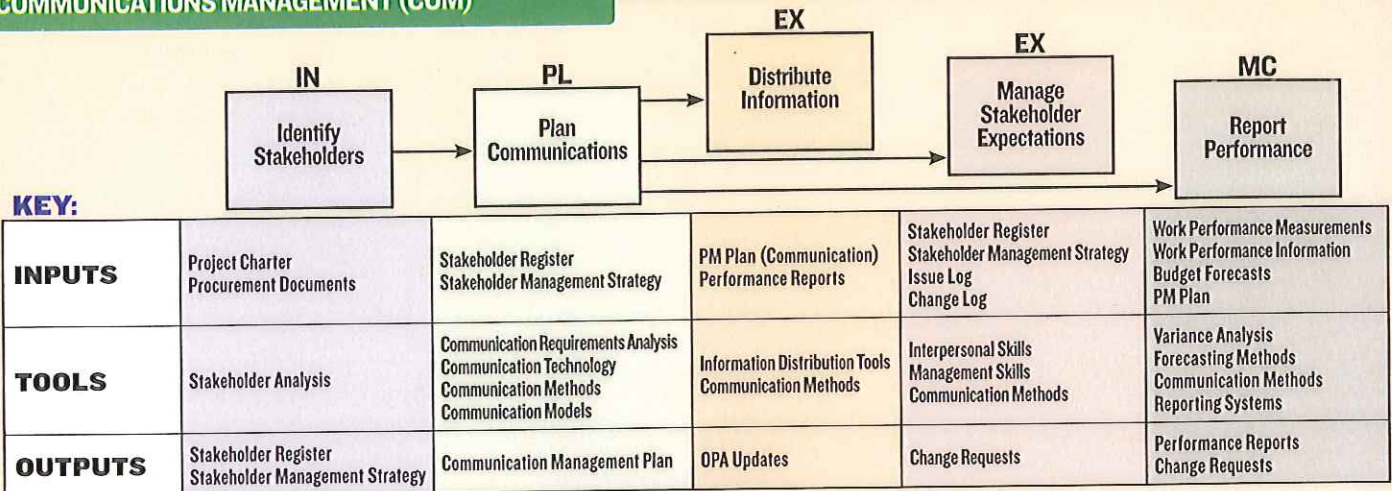
METHODS OF CONFLICT RESOLUTION

Problem-solving*
Collaboration*
Compromise
Forcing
Smoothing
Withdrawal
*Most effective methods

ORIGINS OF PROJECT CONFLICT

1. Schedules,
2. Priorities
3. Resources

COMMUNICATIONS MANAGEMENT (COM)



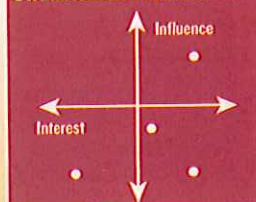
NOTES ABOUT COMMUNICATIONS MANAGEMENT

An effective project manager spends 90% of time communicating, the most important PM skill!

Be proactive and thorough, always distributing accurate information in a timely manner to the right audience

Communications Management Plan contains the following:
Who sends and receives, What is sent, How it is sent, How often it is updated, and a Glossary of terms

STAKEHOLDER ANALYSIS GRID



No. of Communication Channels: $n(n-1) + 2$

Communication Strategies: Interactive, Push, Pull

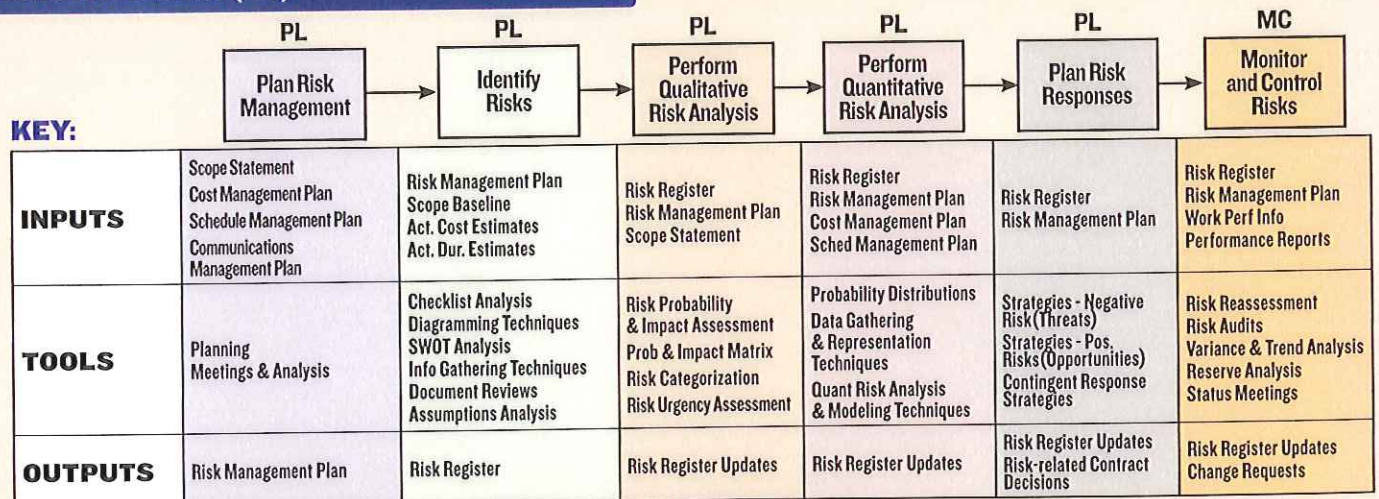
Contract-related Communications: formal and written

Feedback: verbal & nonverbal cues from listener to acknowledge comprehension of the message

Nonverbal Communications: body language, posture, facial expressions, hand motions; most face-to-face communication is nonverbal

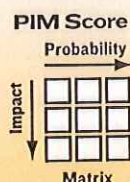
Paralingual: vocal but not verbal - tone of voice, volume, pitch

RISK MANAGEMENT (RIS)



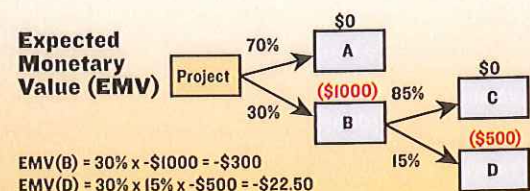
NOTES ABOUT RISK MANAGEMENT

Risks: Uncertainties
can be good or bad; positive (good) risks are opportunities; negative (bad) risks are threats



Opportunity Strategies	Threat Strategies
Share	Transfer
Exploit (100%)	Avoid (0%)
Enhance	Mitigate
Accept: Actively or Passively	

Expected Monetary Value (EMV)





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PROCUREMENT MANAGEMENT (PRO)

KEY:	PL	EX	MC	CL
	Plan Procurements	Conduct Procurements	Administer Procurements	Close Procurements
INPUTS	Scope Baseline Requirements Docs Teaming Agreements Risk-related Contract Decisions Activity Resource Requirements Activity Cost Estimates Cost Performance Baseline	Make-or-buy Decisions Procurement Documents Source Selection Criteria Qualified Seller List Seller Proposals Teaming Agreements	Contract Work Performance Information Procurement Documents Performance Reports	Procurement Documentation
TOOLS	Make-or-buy Analysis Contract Types	Bidder Conferences Proposal Evaluation Techniques Advertising	Contract Change Control System Inspections & Audits Claims Administration	Procurement Audits Negotiated Settlements Records Management System
OUTPUTS	Procurement Management Plan Procurement SOW Make-or-buy Decisions Procurement Documents Source Selection Criteria	Selected Sellers Procurement Contract Award Resource Calendars	Procurement Documentation Change Requests	Closed Procurements

NOTES ABOUT PROCUREMENT MANAGEMENT

CONTRACT TYPES	RISK
Fixed Price	
Firm Fixed Price	Seller
Fixed Price w/ Economic Price Adjustment	Seller
Fixed Price Incentive Fee	Shared
Cost Reimbursable	
Cost Plus Fixed Fee	Buyer
Cost Plus Incentive Fee	Shared
Time & Materials	Buyer

Point of Total Assumption: the cost point at which the seller assumes 100% of the risk of additional cost increases

$PTA = Target\ Cost + ((Ceiling\ Price - Target\ Price) \div Buyer's\ Share)$

Price = Cost + Fee (or Profit)

All contracts and related change requests and communications are formal and written.

PMI® CODE OF ETHICS & PROFESSIONAL CONDUCT (NEITHER A KNOWLEDGE AREA NOR A PROCESS GROUP)

Applies to project managers affiliated with PMI®. Mandatory Standards must be followed; Aspirational Standards are goals.

CHAPTER	MANDATORY STANDARDS	ASPIRATIONAL STANDARDS
RESPONSIBILITY	Take the high road with laws and ethics; Protect whistleblowers	Consider best interests of society, safety, & environment; Take ownership of mistakes; Fulfill commitments
RESPECT	Protect property rights; Negotiate in good faith; Do not abuse others or positions of power	Conduct activities in a professional manner; Resolve conflicts directly; Value other peoples, cultures, & opinions
FAIRNESS	No favoritism or discrimination; Avoid or disclose conflicts of interest	Be transparent in decision processes; Periodically assess your impartiality; Provide equal access to information
HONESTY	Maintain an atmosphere of non-deceit; Never partake in dishonest gain	Promote an environment of truth; Provide accurate information; Ensure truthful communications

> Bribes should always be wrong in test scenarios.

> Always look for the answer that resolves a situation directly, quickly, openly, and fairly.

> Answer with "What should I do?" rather than "What would I do?"

EXAM TIPS

> BEFORE THE EXAM

- Practice listing essential formulas (earned value, PERT and standard deviation, comm channels, and PTA) to list on scratch paper during pre-exam tutorial time.
- Find the testing center the previous day to eliminate anxiety on exam day. Plan to arrive 30 minutes early.
- Do not stay up late and cram the night before the exam; get a full night's rest.
- Prepare a snack to bring if you need to eat something for energy during an exam break.
- Ensure the name on your identification (2 forms, 1 government issued with photo) matches the name on your PMP® application.

> DURING THE EXAM

- ALWAYS FOLLOW THE PROCESS FRAMEWORK:**
Use outputs & inputs to help sequence processes correctly; don't be pressured to bypass a process.
- DON'T TAKE THE EASY WAY OUT:**
Choose the answer that does not include shortcuts or skipping process steps. Do the work!
- ACT DIRECTLY AND SAY WHAT YOU MEAN:**
Be proactive & timely; base decisions on current information; do not pass on the responsibility as PM.
- KNOW THE PROJECT ROLES AND ORG TYPES:**
The PM should always make the decision (do not delegate decision-making).
- DON'T GET STUCK:**
If a question will take several minutes to answer, mark it for review and come back to it later to not run out of exam time.
- FORMULA SHEET:**
After viewing the exam tutorial, use remaining time before exam starts to create a formula sheet on the scratch paper.

7. BUDGET YOUR TIME:

Target an initial pace of 75 questions in 45 minutes; budget 45 minutes for the last 50 questions and the ones skipped.

8. MANAGE YOUR REVIEW:

(1) Did you read the question correctly? (2) Does the answer match the question's logic? (3) Check your math.

9. MANAGE ANXIETY:

If you begin to feel anxious, stop, take a deep breath, hold it for 6 seconds, and release it slowly. Or, take a quick break!

10. READ ALL FOUR ANSWERS AND CHOOSE THE BEST ONE:

They may all be good answers; one is the BEST answer even if it is not the PERFECT answer.

11. READ THE END OF THE QUESTION FIRST:

It may be a simple question, or it may provide better comprehension as the full question is read.

12. USE YOUR INSTINCTS TO ELIMINATE WRONG ANSWERS:

Increase the probability of a correct guess by narrowing the choices.

13. INVESTIGATE FIRST:

When facing a decision, FIRST find out, evaluate, assess, research, determine, understand, or investigate before deciding.